Volume II · Chapter 11 — First Fine-Tunes (PubMedBERT, QLoRA, local MedGemma/Ollama) · Labs

> Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-II-Chapter-11-context.md` for the AI interaction log.

Prerequisites: Ch 7–10.

Learning outcomes — the student can: fine-tune a biomedical BERT for classification; apply QLoRA to fine-tune a small generative LLM on medical QA; run models locally via Ollama; evaluate a fine-tune and write a model card.

Labs (hands-on) — 20 h:

#	Lab	h
11a	Fine-tune PubMedBERT for clinical-question intent classification	5
11b	Prepare a medical QA dataset for generative fine-tuning	3
11c	QLoRA fine-tune a small Llama/GPT-2 on medical QA	6
11d	Run MedGemma locally via Ollama; compare base vs.	3

#	Lab	h
	fine-tuned	
11e	Evaluate the fine-tune (quant + qual) & write a model card	3

Datasets/tools: HF `transformers`/`peft`/`bitsandbytes`, QLoRA, Ollama, MedGemma; GPU workstation. **Assessment:** fine-tuned model + model card (**60%**); eval report (**20%**); quiz (**20%**). **Key decisions:** full vs. PEFT fine-tuning; quantization level vs. quality; base model/size vs. hardware. **References:** plan §13 → *Fine-tuning…*; *Medical LLMs*; *…local inference*. **Hours:** Theory **10** + Lab **20** = **30**.